

Aligning Equations with amsmath

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Introduction

$$\begin{aligned} A &= \frac{\pi r^2}{2} \\ &= \frac{1}{2}\pi r^2 \end{aligned} \tag{1}$$

Writing a Single Equation

$$e^{\pi i} + 1 = 0 \tag{2}$$

The beautiful equation 2 is known as the Euler equation.

Displaying Long Equation

$$\begin{aligned} p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\ - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3 \end{aligned}$$

Splitting and Aligning an Equation

0.1 Aligning Several Equations

$$\begin{array}{lll} 2x - 5y = 8 & & \\ 3x + 9y = -12 & & \\ x = y & w = z & a = b + c \\ 2x = -y & 3w = \frac{1}{2}z & a = b \\ -4 + 5x = 2 + y & w + 2 = -1 + w & ab = cb \end{array}$$